

INVARIANT PLURISUBHARMONIC FUNCTIONS ON NON-COMPACT HERMITIAN SYMMETRIC SPACES.

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ABSTRACT. Let G/K be an irreducible Hermitian symmetric space and let D be a K -invariant domain in G/K . In this paper we characterize several classes of K -invariant plurisubharmonic functions on D in terms of their restrictions to a slice intersecting all K -orbits. As applications we show that K -invariant plurisubharmonic functions on D are necessarily continuous and we reproduce the classification of Stein K -invariant domains in G/K obtained by E. Bedford and J. Dadok in [BeDa91].

1. INTRODUCTION

Let G/K be an irreducible Hermitian symmetric space of rank r . By the polydisk theorem (cf. [Wo72], p.280) the space G/K contains a closed subspace Δ^r , biholomorphic to an r -dimensional polydisk, with the property that $G/K = K \cdot \Delta^r$. If D is a K -invariant domain in G/K , one has $D = K \cdot R$, where $R := D \cap \Delta^r$ is a Reinhardt domain in Δ^r . The polydisk Δ^r and R are invariant under the group $T \ltimes \mathcal{S}_r$, generated by rotations and coordinate permutations.

As the Reinhardt domain R intersects all K -orbits in D , it encodes all information on K -invariant objects in D . In this paper we focus on K -invariant plurisubharmonic functions on D . When D is Stein, we obtain the following characterization of the class $P^{\infty,+}(D)^K$ of smooth, K -invariant, strictly plurisubharmonic functions on D :

$$f \in P^{\infty,+}(D)^K \quad \text{if and only if} \quad f|_R \in P^{\infty,+}(R)^{T \ltimes \mathcal{S}_r},$$

where $f|_R$ is the restriction of f to R . Such result is extended to wider classes of plurisubharmonic functions as follows. Let $P^{\infty}(D)^K$ denote the class of smooth, K -invariant, plurisubharmonic functions and $P^+(D)^K$ (resp. $P(D)^K$) the class of K -invariant, strictly plurisubharmonic (resp. plurisubharmonic) functions on D . One has (Thm. 4.13):

Theorem. *The restriction map $f \rightarrow f|_R$ is a bijection between*

- (i) $P^{\infty,+}(D)^K$ and $P^{\infty,+}(R)^{T \ltimes \mathcal{S}_r}$,
- (ii) $P(D)^K$ and $P(R)^{T \ltimes \mathcal{S}_r}$,
- (iii) $P^{\infty}(D)^K$ and $P^{\infty}(R)^{T \ltimes \mathcal{S}_r}$,
- (iv) $P^+(D)^K$ and $P^+(R)^{T \ltimes \mathcal{S}_r}$.

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As a by-product (Cor. 4.8) we reproduce the classification of Stein K -invariant domains in G/K obtained by Bedford and Dadok in [BeDa91] by a direct computation on some classical cases (see also [FeHu93]).

Corollary. *Let D be a K -invariant domain in G/K .*

- (i) *If G/K is of tube type, then D is Stein if and only if R is Stein and connected.*
- (ii) *If G/K is not of tube type, then D is Stein if and only if R is Stein and complete. In particular R contains the origin and is connected.*

Let $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ be a Cartan decomposition of the Lie algebra $\mathfrak{g}$ of G , let $\mathfrak{a}$ be a maximal abelian subspace of $\mathfrak{p}$, with Weyl group W , and let $G = K \exp \mathfrak{a} K$ be the corresponding decomposition of G . Every K -invariant domain D in G/K is uniquely determined by a W -invariant domain $\mathcal{D}_{\mathfrak{a}}$ in $\mathfrak{a}$ by

$$\mathcal{D}_{\mathfrak{a}} \rightarrow K \exp \mathcal{D}_{\mathfrak{a}} K / K = D.$$

Moreover, to every smooth K -invariant function f on D there corresponds a unique smooth W -invariant function $\tilde{f}$ on $\mathcal{D}_{\mathfrak{a}}$, defined by

$$\tilde{f}(H) := f(\exp(H)K), \quad H \in \mathcal{D}_{\mathfrak{a}},$$

(cf. [Fle78], [Dad82]). The proof of the above results is carried out as follows. As a first step we explicitly compute the Levi form of f in terms of the first and second derivatives of $\tilde{f}$. This is achieved in Proposition 3.1 by means of a fine decomposition of the tangent bundle of D , induced by the restricted root decomposition of $\mathfrak{g}$ (Rem. 2.3), and a simple pluripotential argument which enable us to maximally exploit the involved symmetries.

The Levi form computation leads to the key ingredient of our results. Namely, the following characterization of smooth K -invariant strictly plurisubharmonic functions on a Stein K -invariant domain D (Thm. 4.5):

$$f \in P^{\infty,+}(D)^K \quad \text{if and only if} \quad \tilde{f} \in \text{LogConv}^{\infty,+}(\mathcal{D}_{\mathfrak{a}})^W,$$

where the latter class consists of smooth W -invariant functions on $\mathcal{D}_{\mathfrak{a}}$ satisfying an appropriate differential positivity condition. We also show that $\tilde{f}$ belongs to $\text{LogConv}^{\infty,+}(\mathcal{D}_{\mathfrak{a}})^W$ if and only if the corresponding $T \times \mathcal{S}_r$ -invariant function on R is strictly plurisubharmonic (Prop. 4.1). This fact, which may be of independent interest in the context of Reinhardt domains, implies (i) in the above theorem. When extending the above characterization to the non-smooth setting (Thm. 4.12), it turns out that K -invariant plurisubharmonic functions on D are necessarily continuous.

Finally, in the appendix we explicitly determine a K -invariant potential of the Killing form on G/K (Prop. 5.1) and we observe that, up to an additive constant, it coincides with the logarithm of the Bergman kernel function.

We remark that our methods require no classification results, nor any distinction between classical and exceptional cases.

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2. PRELIMINARIES

Let $\mathfrak{g}$ be a non-compact semisimple Lie algebra and let $\mathfrak{k}$ be a maximal compact subalgebra of $\mathfrak{g}$. Let $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ be the Cartan decomposition of $\mathfrak{g}$ with respect to $\mathfrak{k}$, with Cartan involution θ . Let $\mathfrak{a}$ be a maximal abelian subspace in $\mathfrak{p}$. The dimension r of $\mathfrak{a}$ is by definition the *rank* of G/K . Let $\mathfrak{g} = \mathfrak{m} \oplus \mathfrak{a} \oplus \bigoplus_{\alpha \in \Sigma} \mathfrak{g}^\alpha$ be the restricted root decomposition of $\mathfrak{g}$, where $\mathfrak{m}$ is the centralizer of $\mathfrak{a}$ in $\mathfrak{k}$, the joint eigenspace $\mathfrak{g}^\alpha = \{X \in \mathfrak{g} \mid [H, X] = \alpha(H)X, \text{ for all } H \in \mathfrak{a}\}$ is the α -restricted root space and the restricted root system Σ consists of those $\alpha \in \mathfrak{a}^*$ for which $\mathfrak{g}^\alpha \neq \{0\}$. Denote by $B(\cdot, \cdot)$ the Killing form of $\mathfrak{g}$, as well as its holomorphic extension to $\mathfrak{g}^\mathbb{C}$ (which coincides with the Killing form of $\mathfrak{g}^\mathbb{C}$). Denote by W the Weyl group of $\mathfrak{a}$, i.e. the quotient of the normalizer over the centralizer of $\mathfrak{a}$ in K . In the Hermitian case W acts on $\mathfrak{a}$ by signed permutations.

For $\alpha \in \Sigma$, consider the θ -stable space $\mathfrak{g}[\alpha] := \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}$, and denote by $\mathfrak{k}[\alpha]$ and $\mathfrak{p}[\alpha]$ the projections of $\mathfrak{g}[\alpha]$ along $\mathfrak{p}$ and $\mathfrak{k}$, respectively. Let Σ^+ be a choice of positive roots in Σ . Then

$$\mathfrak{k} = \mathfrak{m} \oplus \bigoplus_{\alpha \in \Sigma^+} \mathfrak{k}[\alpha] \quad \text{and} \quad \mathfrak{p} = \mathfrak{a} \oplus \bigoplus_{\alpha \in \Sigma^+} \mathfrak{p}[\alpha] \quad (1)$$

are B -orthogonal decompositions of $\mathfrak{k}$ and $\mathfrak{p}$, respectively. The next lemma is an easy exercise.

Lemma 2.1. *Every element X in $\mathfrak{p}$ decomposes in a unique way as*

$$X_{\mathfrak{a}} + \sum_{\alpha \in \Sigma^+} P^\alpha,$$

where $X_{\mathfrak{a}} \in \mathfrak{a}$ and $P^\alpha \in \mathfrak{p}[\alpha]$. The vector P^α can be written uniquely as $P^\alpha = X^\alpha - \theta X^\alpha$, where X^α is the component of X in the root space $\mathfrak{g}^\alpha$. Moreover, $[H, P^\alpha] = \alpha(H)K^\alpha$, where K^α is the element in $\mathfrak{k}[\alpha]$ uniquely defined by $K^\alpha = X^\alpha + \theta X^\alpha$.

The restricted root system of a Lie algebra $\mathfrak{g}$ of Hermitian type is either of type C_r (if G/K is of tube type) or of type BC_r (if G/K is not of tube type), i.e. there exists a basis $\{e_1, \dots, e_r\}$ of $\mathfrak{a}^*$ for which

$$\Sigma^+ = \{2e_j, 1 \leq j \leq r, e_k \pm e_l, 1 \leq k < l \leq r\}, \quad \text{for type } C_r,$$

$$\Sigma^+ = \{e_j, 2e_j, 1 \leq j \leq r, e_k \pm e_l, 1 \leq k < l \leq r\}, \quad \text{for type } BC_r.$$

With the above choice of a positive system Σ^+ , the roots

$$2e_1, \dots, 2e_r$$

form a maximal set of long strongly orthogonal positive restricted roots (i.e. such that $2e_k \pm 2e_l \notin \Sigma$, for $k \neq l$).

Denote by I_0 the G -invariant complex structure of G/K . For every $j = 1, \dots, r$, the root space $\mathfrak{g}^{2e_j}$ is one-dimensional. Choose generators $E^j \in \mathfrak{g}^{2e_j}$ such that the $\mathfrak{sl}(2)$ -triples $\{E^j, \theta E^j, A_j := [\theta E^j, E^j]\}$ are normalized as follows

$$[A_j, E^j] = 2E^j, \quad \text{for } j = 1, \dots, r. \quad (2)$$

We also assume that $I_0(E^j - \theta E^j) = A_j$ (see [GeIa13], Def. 2.1). By the strong orthogonality of $2e_1, \dots, 2e_r$, the vectors $A_1, \dots, A_r$ form a B -orthogonal basis

of $\mathfrak{a}$, dual to the basis $e_1, \dots, e_r$ of $\mathfrak{a}^*$, and the associated $\mathfrak{sl}(2)$ -triples pairwise commute. For $j = 1, \dots, r$, define

$$K^j := E^j + \theta E^j \quad \text{and} \quad P^j := E^j - \theta E^j. \quad (3)$$

On $\mathfrak{p} \cong T_{eK}G/K$ the complex structure I_0 coincides with the adjoint action of the element $Z_0 \in Z(\mathfrak{k})$ given by

$$Z_0 = S_0 + \frac{1}{2} \sum_{j=1}^r K^j, \quad (4)$$

for some element S_0 in a Cartan subalgebra $\mathfrak{s}$ of $\mathfrak{m}$. One has $S_0 = 0$ in the tube case (see [GeIa13], Lem. 2.2). The complex structure I_0 permutes the blocks of the decomposition (1) of $\mathfrak{p}$ (cf. [RoVe73]), namely

$$I_0 \mathfrak{a} = \bigoplus_{j=1}^r \mathfrak{p}[2e_j], \quad I_0 \mathfrak{p}[e_j + e_l] = \mathfrak{p}[e_j - e_l], \quad I_0 \mathfrak{p}[e_j] = \mathfrak{p}[e_j]. \quad (5)$$

In order to state the next result, we need to recall a few more facts. Let $\mathfrak{g}^{\mathbb{C}} = \mathfrak{h}^{\mathbb{C}} \oplus \bigoplus_{\mu \in \Delta} \mathfrak{g}^{\mu}$ be the root decomposition of $\mathfrak{g}^{\mathbb{C}}$ with respect to the maximally split Cartan subalgebra $\mathfrak{h} = \mathfrak{s} \oplus \mathfrak{a}$ of $\mathfrak{g}$. Let σ be the conjugation of $\mathfrak{g}^{\mathbb{C}}$ with respect to $\mathfrak{g}$. Let θ denote also the $\mathbb{C}$ -linear extension of θ to $\mathfrak{g}^{\mathbb{C}}$. One has $\theta\sigma = \sigma\theta$. Write $\bar{Z} := \sigma Z$, for $Z \in \mathfrak{g}^{\mathbb{C}}$. As σ and θ stabilize $\mathfrak{h}$, they induce actions on Δ , defined by $\bar{\lambda}(H) := \overline{\lambda(H)}$ and $\theta\lambda(H) := \lambda(\theta(H))$, for $H \in \mathfrak{h}$, respectively. Fix a positive root system Δ^+ compatible with Σ^+ , meaning that $\lambda|_{\mathfrak{a}} = \text{Re}(\lambda) \in \Sigma^+$ implies $\lambda \in \Delta^+$. Then $\sigma\Delta^+ = \Delta^+$.

The next lemma gives a more detailed description of the complex structure I_0 on $\mathfrak{p}$.

Lemma 2.2.

(a) For $j = 1, \dots, r$, let A_j and P^j be as in (2) and (3). One has $I_0 P^j = A_j$ and $I_0 A_j = -P^j$.

(b) Let $P = X - \theta X \in \mathfrak{p}[e_j + e_l]$, where $X = Z^{\mu} + \bar{Z}^{\mu}$, with $Z^{\mu} \in \mathfrak{g}^{\mu}$ and $\mu \in \Delta^+$ is a root satisfying $\text{Re}(\mu) = e_j + e_l$. If $\bar{\mu} = \mu$, we may assume $Z^{\mu} = \bar{Z}^{\mu}$ and set $X = Z^{\mu}$. Then $I_0 P = Y - \theta Y$, where $Y = [Z_0, X] = \frac{1}{2}[K^j + K^l, X] \in \mathfrak{g}^{e_j - e_l} \oplus \mathfrak{g}^{\theta(e_j - e_l)}$.

(c) Let $P = X - \theta X \in \mathfrak{p}[e_j]$, where $X = Z^{\mu} + \bar{Z}^{\mu}$, $Z^{\mu} \in \mathfrak{g}^{\mu}$ and μ is a root in Δ^+ satisfying $\text{Re}(\mu) = e_j$ (as $\dim \mathfrak{p}[e_j]$ is even, necessarily $\bar{\mu} \neq \mu$). Then $I_0 P = Y - \theta Y$, where $Y = i(Z^{\mu} - \bar{Z}^{\mu}) \in \mathfrak{g}^{e_j}$.

Proof. (a) follows by definition from (2) and (3).

Observe that $Z_0 \in Z(\mathfrak{k})$ implies

$$[Z_0, X] = -[Z_0, \theta X], \quad \text{for every } X \in \mathfrak{g}. \quad (6)$$

(b) By (4), (5) and the fact that $[S_0, \mathfrak{g}^{\mu}] \subset \mathfrak{g}^{\mu}$, for every $\mu \in \Delta$, the action of S_0 is necessarily trivial on $\mathfrak{p}[e_j + e_l]$. Moreover, if $X \in \mathfrak{g}^{e_j + e_l}$, then $[K^i, X \pm \theta X] = 0$, for all $i \neq j, l$, implying that $[Z_0, X \pm \theta X] = \frac{1}{2}[K^j + K^l, X \pm \theta X]$. Note that $[K^l, Z^{\mu}] \in \mathfrak{g}^{\lambda}$ and $[K^j, Z^{\mu}] \in \mathfrak{g}^{\theta\lambda}$, where $\lambda \in \Delta$ is the root with real part $e_j - e_l$ and the same imaginary part as μ . Then, by equating terms in

the same root spaces in (6), one obtains the relations which guarantee that $Y = [Z_0, X] \in \mathfrak{g}^{e_j - e_l} \oplus \mathfrak{g}^{\theta(e_j - e_l)}$. Namely

$$\begin{aligned} [K^l, Z^\mu] &= -[K^j, \theta Z^\mu] \in \mathfrak{g}^\lambda & [K^l, \bar{Z}^\mu] &= -[K^j, \theta \bar{Z}^\mu] \in \mathfrak{g}^{\bar{\lambda}}, \\ [K^j, Z^\mu] &= -[K^l, \theta Z^\mu] \in \mathfrak{g}^{\theta\lambda} & [K^j, \bar{Z}^\mu] &= -[K^l, \theta \bar{Z}^\mu] \in \mathfrak{g}^{\theta\bar{\lambda}}. \end{aligned}$$

(c) If $X \in \mathfrak{g}^{e_j}$, then $[K^l, X \pm \theta X] = 0$, for all $l \neq j$, implying that $[Z_0, X \pm \theta X] = [\frac{1}{2}K^j + S_0, X \pm \theta X]$. From (6) one obtains

$$\frac{1}{2}[K^j, X] + [S_0, X] = -\frac{1}{2}[K^j, \theta X] - [S_0, \theta X],$$

and, by equating terms in the same root spaces, one obtains the relations

$$\begin{aligned} [S_0, Z^\mu] &= -\frac{1}{2}[K^j, \theta Z^\mu] \in \mathfrak{g}^\mu & [S_0, \theta Z^\mu] &= -\frac{1}{2}[K^j, Z^\mu] \in \mathfrak{g}^{\theta\mu} \\ [S_0, \bar{Z}^\mu] &= -\frac{1}{2}[K^j, \theta \bar{Z}^\mu] \in \mathfrak{g}^{\bar{\mu}} & [S_0, \theta \bar{Z}^\mu] &= -\frac{1}{2}[K^j, \bar{Z}^\mu] \in \mathfrak{g}^{\theta\bar{\mu}}, \end{aligned}$$

which imply

$$[Z_0, X - \theta X] = 2(-[S_0, \theta Z^\mu] - [S_0, \theta \bar{Z}^\mu] + [S_0, Z^\mu] + [S_0, \bar{Z}^\mu]).$$

As $\mu(S_0) =: i\mu_0 \in i\mathbb{R}$, the above expression becomes

$$2\mu_0 i(Z^\mu - \bar{Z}^\mu - \theta(Z^\mu - \bar{Z}^\mu)).$$

From $I_0^2 = -Id$, one obtains $\mu_0 = \pm \frac{1}{2}$. Depending on the value μ_0 , the pairs of roots $\mu, \bar{\mu}$ can be relabelled so that $I_0 P$, has the desired expression. $\square$

Remark 2.3. In view of Lemma 2.2, one can choose a I_0 -stable basis of $\mathfrak{p}$, compatible with the decomposition (1).

(a) A basis of $\mathfrak{a} \oplus \bigoplus_j \mathfrak{p}[2e_j]$: take the elements $A_j, P_j = -I_0 A_j$, for $j = 1, \dots, r$, normalized as in (2) and (3);

(b) A basis of $\mathfrak{p}[e_j + e_l] \oplus \mathfrak{p}[e_j - e_l]$: take 4-tuples of elements $P, P', I_0 P, I_0 P'$, parametrized by the pairs of roots $\mu \neq \bar{\mu} \in \Delta^+$ satisfying $\text{Re}(\mu) = e_j + e_l$ (with no repetition). More precisely, $P = X - \theta X$ and $P' = X' - \theta X'$, where $X = Z^\mu + \bar{Z}^\mu$, $X' = i(Z^\mu - \bar{Z}^\mu)$, and Z^μ is a root vector in $\mathfrak{g}^\mu$. For $\mu = \bar{\mu}$, we may assume $Z^\mu = \bar{Z}^\mu$. Then take the pair $P, I_0 P$.

(c) A basis of $\mathfrak{p}[e_j]$ (non-tube case): take pairs of elements $P, I_0 P$, parametrized by the pairs of roots $\mu \neq \bar{\mu} \in \Delta^+$ satisfying $\text{Re}(\mu) = e_j$ (with no repetition). More precisely, $P = X - \theta X$, where $X = Z^\mu + \bar{Z}^\mu$, and Z^μ is a root vector in $\mathfrak{g}^\mu$.

Lemma 2.4. Let $\mu \in \Delta^+$ be a root satisfying $\text{Re}(\mu) = e_j + e_l$ and let Z^μ a root vector in $\mathfrak{g}^\mu$. Let $X = Z^\mu + \bar{Z}^\mu \in \mathfrak{g}^{e_j + e_l}$ and $Y = [Z_0, X]$. Then

- (a) $[Y, X] + \theta[Y, X] = r_1 K^j + s_1 K^l$, for $r_1, s_1 \in \mathbb{R}$;
- (b) $[Y, \theta X] + \theta[Y, \theta X] = r_2 K^j + s_2 K^l$, for $r_2, s_2 \in \mathbb{R}$.

If $\bar{\mu} \neq \mu$, let $X' = i(Z^\mu - \bar{Z}^\mu)$ and $Y' = [Z_0, X']$. Then

- (c) $[Y', X] + \theta[Y', X] = [Y', \theta X] + \theta[Y', \theta X] = 0$.

Let μ be a root in Δ^+ , with $\operatorname{Re}(\mu) = e_j$ (non-tube case) and let Z^μ be a root vector in $\mathfrak{g}^\mu$. Let $X = Z^\mu + \overline{Z}^\mu$ and $Y = [Z_0, X] = i(Z^\mu - \overline{Z}^\mu)$. Then

(d) $[Y, X] + \theta[Y, X] = tK^j$, for $t \in \mathbb{R}$,

(e) $[Y, \theta X] + \theta[Y, \theta X] \in \mathfrak{m}$.

Proof. (a) By Lemma 2.2 (b), one has $Y = \frac{1}{2}[K^j + K^l, X] \in \mathfrak{g}^{e_j - e_l} \oplus \mathfrak{g}^{\theta(e_j - e_l)}$. Moreover $[Y, X] = \frac{1}{2}[[\theta E^j, X], X] + \frac{1}{2}[[\theta E^l, X], X] \in \mathfrak{g}^{2e_l} \oplus \mathfrak{g}^{2e_j}$. Since such roots spaces are 1-dimensional, then

$$[Y, X] + \theta[Y, X] = r_1 K^j + s_1 K^l, \quad \text{for some } r_1, s_1 \in \mathbb{R}.$$

(b) Similarly, $[Y, \theta X] = \frac{1}{2}[[\theta E^j, X], \theta X] + \frac{1}{2}[[\theta E^l, X], \theta X] \in \mathfrak{g}^{-2e_j} \oplus \mathfrak{g}^{-2e_l}$, and

$$[Y, \theta X] + \theta[Y, \theta X] = s_1 K^j + s_2 K^l, \quad \text{for some } s_1, s_2 \in \mathbb{R}.$$

(c) One has

$$\begin{aligned} [Y', X] &= [[Z_0, iZ^\mu - i\overline{Z}^\mu], Z^\mu + \overline{Z}^\mu] \\ &= i[[Z_0, Z^\mu], Z^\mu] + i[[Z_0, Z^\mu], \overline{Z}^\mu] - i[[Z_0, \overline{Z}^\mu], Z^\mu] - i[[Z_0, \overline{Z}^\mu], \overline{Z}^\mu]. \end{aligned}$$

The first and the fourth terms of the above expression sum up to zero: if $\bar{\mu} \neq \mu$, they are both zero because otherwise there would exist a root in Δ^+ with real part equal to $2e_j$ and non-zero imaginary part; if $\bar{\mu} = \mu$, such terms are opposite to each other. The second and the third term sum up to zero by the Jacobi identity and the fact that $[Z_0, [Z^\mu, \overline{Z}^\mu]] = 0$. One has

$$\begin{aligned} [Y', \theta X] &= [[Z_0, iZ^\mu - i\overline{Z}^\mu], \theta Z^\mu + \theta \overline{Z}^\mu] \\ &= i[[Z_0, Z^\mu], \theta Z^\mu] + i[[Z_0, Z^\mu], \theta \overline{Z}^\mu] - i[[Z_0, \overline{Z}^\mu], \theta Z^\mu] - i[[Z_0, \overline{Z}^\mu], \theta \overline{Z}^\mu]. \end{aligned}$$

The above expression is automatically zero, if $\bar{\mu} = \mu$. So let's assume that $\bar{\mu} \neq \mu$. Arguing as in the previous case, one has that the sum of the first and the fourth terms is equal to zero. The second and the third terms sum up to $2\operatorname{Im}([Z_0, Z^\mu], \theta \overline{Z}^\mu)$. Then

$$\begin{aligned} [Y', \theta X] + \theta[Y', \theta X] &= 2\operatorname{Im}([Z_0, Z^\mu], \theta \overline{Z}^\mu) + \theta[[Z_0, Z^\mu], \theta \overline{Z}^\mu] = \\ &= 2\operatorname{Im}([Z_0, Z^\mu], \theta \overline{Z}^\mu) + [[Z_0, \theta Z^\mu], \overline{Z}^\mu]. \end{aligned} \tag{7}$$

By the Jacobi identity

$$[[Z_0, \theta Z^\mu], \overline{Z}^\mu] = [[Z_0, \overline{Z}^\mu], \theta Z^\mu] + [Z_0, [\theta Z^\mu, \overline{Z}^\mu]].$$

Observe that $[Z^\mu, \theta \overline{Z}^\mu] \in \mathfrak{a} \oplus i\mathfrak{s}$ and therefore $[Z_0, [\theta Z^\mu, \overline{Z}^\mu]] \in \mathfrak{p}$. It follows that the expression in (7) reduces to

$$2\operatorname{Im}([Z_0, Z^\mu], \theta \overline{Z}^\mu) + [[Z_0, \overline{Z}^\mu], \theta Z^\mu] = 0,$$

as desired.

(d) Since $[Y, X] \in \mathfrak{g}^{2e_j}$

$$[Y, X] + \theta[Y, X] = tK^j, \quad \text{for some } t \in \mathbb{R}.$$

(e) Since $[Y, X] \in \mathfrak{g}^0$,

$$[Y, X] + \theta[Y, X] \in \mathfrak{m}.$$

□

Next, we recall a fact which will be used to compute the Levi form of an arbitrary smooth K -invariant function on G/K . For $X \in \mathfrak{k}$, denote by $\tilde{X}$ the vector field induced on G/K by the K -action

$$\tilde{X}_z := \left. \frac{d}{ds} \right|_{s=0} \exp sX \cdot z, \quad z \in G/K. \quad (8)$$

Given a smooth K -invariant function $f: G/K \rightarrow \mathbb{R}$, set $d_{I_0}^c \rho := d\rho \circ I_0$, so that $2i\partial\bar{\partial}_{I_0} f = -dd_{I_0}^c f$. Moreover, for $X \in \mathfrak{k}$, define $\mu^X: G/K \rightarrow \mathbb{R}$ by $\mu^X(z) := d_{I_0}^c f(\tilde{X}_z)$. Then

$$d\mu^X = -\iota_{\tilde{X}} dd_{I_0}^c f. \quad (9)$$

The above identity was proved in [HeSc07], Lemma 7.1. Indeed their argument needs not the plurisubharmonicity of f .

If the function f is strictly plurisubharmonic, then $-dd_{I_0}^c f$ is a K -invariant Kähler form and the map $\mu: G/K \rightarrow \mathfrak{k}^*$, defined by

$$\mu(z)(X) = d_{I_0}^c f(\tilde{X}_z),$$

for $X \in \mathfrak{k}$, is a moment map. It is referred to as the moment map associated with f .

We conclude the preliminaries with a lemma needed in the next section. Let Δ be the unit disc in $\mathbb{C}$. Consider the $(T \ltimes \mathcal{S}_2)$ -action on Δ^2 , where $T = (S^1)^2$ acts by rotations and $\mathcal{S}_2$ is the group of coordinate permutations. Let $W_{\mathbb{R}^2} = (\mathbb{Z}_2) \ltimes \mathcal{S}_2$ be the group acting on $\mathbb{R}^2$ by signed permutations.

Lemma 2.5. *Let $f: \Delta^2 \rightarrow \mathbb{R}$ be a smooth $T \ltimes \mathcal{S}_2$ -invariant strictly plurisubharmonic function and let r, s be real numbers. Consider the $W_{\mathbb{R}^2}$ -invariant function $\tilde{f}: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $\tilde{f}(a_1, a_2) = f(\tanh a_1, \tanh a_2)$ and define*

$$G_{\tilde{f}}(a_1, a_2) := \frac{r \sinh(2a_1) \frac{\partial \tilde{f}}{\partial a_1}(x, y) - s \sinh(2a_2) \frac{\partial \tilde{f}}{\partial a_2}(a_1, a_2)}{\sinh^2 a_1 - \sinh^2 a_2}.$$

Then

(i) $\frac{\partial \tilde{f}}{\partial a_1}(a_1, a_2) > 0$, for every $a_1 > 0$, and $\frac{\partial \tilde{f}}{\partial a_1}(a_1, a_2) < 0$, for every $a_1 < 0$.

In particular $\frac{\partial \tilde{f}}{\partial a_1}(0, a_2) = 0$, for every real a_2 .

(ii) $\frac{\partial \tilde{f}}{\partial a_2}(a_1, a_2) = \frac{\partial \tilde{f}}{\partial a_1}(a_2, a_1)$. In particular $\frac{\partial \tilde{f}}{\partial a_2}(a_1, 0) = 0$, for every real a_1 .

(iii) If $G_{\tilde{f}}$ extends continuously on $\mathbb{R}^2$ to a strictly positive function, then $r = s > 0$. In particular $G_{\tilde{f}}(a_1, a_2)$ is $W_{\mathbb{R}^2}$ -invariant as well.

Proof. (i) For $a_1 > 0$ let $s_1 \in (-\infty, 0)$ be such that $\tanh a_1 = e^{s_1}$. Since f is T -invariant and strictly plurisubharmonic, the function $s_1 \rightarrow f(e^{s_1}, a_2)$ is strictly convex. Moreover, the limit $\lim_{s_1 \rightarrow -\infty} f(e^{s_1}, a_2) = f(0, a_2)$ is finite. Hence the function is strictly increasing and its derivative $e^{s_1} \frac{\partial f}{\partial a_1}(e^{s_1}, a_2)$ is strictly positive. As

$$\frac{\partial \tilde{f}}{\partial a_1}(a_1, a_2) = \frac{1}{\cosh^2 a_1} \frac{\partial f}{\partial x}(\tanh a_1, a_2),$$

and $\tanh a_1 = e^{s_1}$, statement (i) follows.

(ii) The $\mathcal{S}_2$ -invariance of $\tilde{f}$ implies that

$$\lim_{\varepsilon \rightarrow 0} \frac{\tilde{f}(a_1, a_2 + \varepsilon)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{\tilde{f}(a_2 + \varepsilon, a_1)}{\varepsilon},$$

and (ii) follows.

(iii) Let $a_1 > 0$. From (ii) it follows that $G_{\tilde{f}}(a_1, 0) = \frac{r \sinh(2a_1) \frac{\partial \tilde{f}}{\partial a_1}(a_1, 0)}{\sinh^2 a_1}$. Since such quantity is assumed to be strictly positive and $\frac{\partial \tilde{f}}{\partial a_1}(a_1, 0) > 0$, it follows that $r > 0$. By choosing $a_1 = 0$ and $a_2 > 0$, one obtains that $s > 0$.

Next we show that $r = s$. For $a_1 > a_2 > 0$ one has $\sinh^2 a_1 - \sinh^2 a_2 > 0$. Then the positivity of $G_{\tilde{f}}(a_1, a_2)$ implies that

$$\frac{r}{s} > \frac{\sinh(2a_2) \frac{\partial \tilde{f}}{\partial a_2}(a_1, a_2)}{\sinh(2a_1) \frac{\partial \tilde{f}}{\partial a_1}(a_1, a_2)}.$$

Consequently, for a_1 converging to a fixed $a_2 > 0$, statements (i) and (ii) imply $\frac{r}{s} \geq 1$. An analogous argument, with $0 < a_1 < a_2$, implies $\frac{r}{s} \leq 1$. As a consequence, $\frac{r}{s} = 1$. $\square$

3. THE LEVI FORM OF A K -INVARIANT FUNCTION

Let G/K an irreducible non-compact Hermitian symmetric space of rank r . From the decomposition $G = K \exp \mathfrak{a} K$ one obtains a bijective correspondence between W -invariant domains in $\mathfrak{a}$ and K -invariant domains in G/K , namely

$$\mathcal{D}_{\mathfrak{a}} \rightarrow D := K \exp \mathcal{D}_{\mathfrak{a}} K / K. \quad (10)$$

In addition, every K -invariant function $f : D \rightarrow \mathbb{R}$ is uniquely determined by the W -invariant function $\tilde{f} : \mathcal{D}_{\mathfrak{a}} \rightarrow \mathbb{R}$, given by

$$\tilde{f}(H) = f(\exp(H)K). \quad (11)$$

The goal of this section is express the Hermitian form $h_f(\cdot, \cdot) := -dd^c f(\cdot, I_0 \cdot)$ of a smooth K -invariant function f on a K -invariant domain $D \subset G/K$ in terms of the first and second derivatives of the corresponding $\tilde{f}$ on $\mathcal{D}_{\mathfrak{a}}$. This will enable us to characterize smooth K -invariant strictly plurisubharmonic functions on a Stein K -invariant domain D in G/K by an appropriate differential positivity condition on the corresponding functions on $\mathcal{D}_{\mathfrak{a}}$ (see Thm. 4.5 and Cor. 4.6). As f is K -invariant, also $-dd^c f(\cdot, I_0 \cdot)$ is K -invariant. Therefore it will be sufficient to carry out the computation along the slice $\exp \mathfrak{a} K$, which meets all K -orbits.

For $z = aK$, with $a = \exp(H)$ and $H \in \mathfrak{a}$, one has

$$\tilde{X}_z = a_* F_a X, \quad a_* X = \widetilde{F_a^{-1} X_z}, \quad (12)$$

where $F_a : \mathfrak{p} \rightarrow \mathfrak{p}$ is the map given by $F_a := \pi_{\#} \circ \text{Ad}_{a^{-1}}|_{\mathfrak{p}}$, and $\pi_{\#} : \mathfrak{g} \rightarrow \mathfrak{p}$ is the linear projection along $\mathfrak{k}$. In particular one can verify that

$$\tilde{K}_z = -a_* \sinh \alpha(H) P, \quad (13)$$

for $P = X^{\alpha} - \theta X^{\alpha} \in \mathfrak{p}[\alpha]$ and $K = X^{\alpha} + \theta X^{\alpha} \in \mathfrak{k}[\alpha]$, with $\alpha \in \Sigma^{+}$.

Denote by $a_1, \dots, a_r$ the coordinates induced on $\mathfrak{a}$ by the basis $A_1, \dots, A_r$ of $\mathfrak{a}$ (cf. Rem. 2.3(a)).

Proposition 3.1. *Let $D \subset G/K$ be a K -invariant domain. Let $f : D \rightarrow \mathbb{R}$ be a smooth K -invariant function. Fix $a = \exp H$, with $H = \sum_j a_j A_j \in \mathcal{D}_{\mathfrak{a}}$. Then, in the basis of $\mathfrak{p}$ defined in Remark 2.3, the Hermitian form h_f at $z = aK \in D$ is as follows.*

- (i) *The spaces $a_*\mathfrak{a}$, $a_*I_0\mathfrak{a}$, $a_*\mathfrak{p}[e_j + e_l]$, $a_*\mathfrak{p}[e_j - e_l]$ and $a_*\mathfrak{p}[e_j]$ are pairwise h_f -orthogonal.*

As the form h_f is I_0 -invariant, it is determined by its restrictions to the blocks $a_\mathfrak{a}$, $a_*\mathfrak{p}[e_j + e_l]$ and $a_*\mathfrak{p}[e_j]$. The non-zero entries of h_f on each of these blocks are given as follows.*

- (ii) *For $A_j, A_l \in \mathfrak{a}$ one has*

$$h_f(a_*A_j, a_*A_l) = 2 \coth(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(H) \delta_{jl} + \frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l}(H);$$

- (iii) *For P, P' as in Remark 2.3(b) one has*

$$\begin{aligned} h_f(a_*P, a_*P) &= h_f(a_*P', a_*P') = \\ &= \frac{B(P, P)}{b} \frac{1}{\sinh(a_j + a_l) \sinh(a_j - a_l)} \left(\sinh(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(H) - \sinh(2a_l) \frac{\partial \tilde{f}}{\partial a_l}(H) \right), \end{aligned}$$

where $b := B(A_1, A_1) = \dots = B(A_r, A_r)$. In particular, with respect to the basis of $a_\mathfrak{p}[e_j + e_l]$ defined in Rem. 2.3(b), the form h_f is diagonal.*

- (iv) *(non-tube case) For $P \in \mathfrak{p}[e_j]$, as in Remark 2.3 (c), one has*

$$h_f(a_*P, a_*P) = \frac{B(P, P)}{b} \coth(a_j) \frac{\partial \tilde{f}}{\partial a_j}(H).$$

In particular, with respect to the basis of $a_\mathfrak{p}[e_j]$ defined in Rem. 2.3(c), the form h_f is diagonal.*

Proof. In order to exploit the relation (9), we first compute $d^c f(\tilde{X}_z)$, for $X \in \mathfrak{k}$ and $z \in G/K$. By the K -invariance of f and of I_0 one has

$$d^c f(\tilde{X}_{k \cdot z}) = d^c f(\widetilde{\text{Ad}_{k^{-1}} X_z}), \quad (14)$$

for every $z \in G/K$ and $k \in K$. Thus it is sufficient to carry out the computation for $z = aK$. We first assume that $\alpha(H) \neq 0$ for all $\alpha \in \Sigma$, and later obtain the desired result by passing to the limit for H approaching the hyperplanes defined by $\{\alpha = 0\}$.

Recall the decomposition of $\mathfrak{k}$ given in (1). For all $M \in \mathfrak{m}$, one has $\tilde{M}_z = 0$, and therefore

$$d^c f(\tilde{M}_z) = 0. \quad (15)$$

For $K = X^\alpha + \theta X^\alpha$ in $\mathfrak{k}[\alpha]$, with $\alpha \neq 2e_1, \dots, 2e_r$, set $P = X^\alpha - \theta X^\alpha$ in $\mathfrak{p}[\alpha]$. Then $I_0 P = Y^\beta - \theta Y^\beta$ in $\mathfrak{p}[\beta]$, for some $\beta \neq 0$. Set $C = Y^\beta + \theta Y^\beta$ in $\mathfrak{k}[\beta]$. Then, by (13) and the K -invariance of f , one has

$$\begin{aligned} d^c f(\tilde{K}_z) &= -df(I_0 a_* \sinh \alpha(H) P) = -df(a_* \sinh \alpha(H) I_0 P) \\ &= df\left(\frac{\sinh \alpha(H)}{\sinh \beta(H)} \tilde{C}_z\right) = 0. \end{aligned} \quad (16)$$

Finally, in the case of $K^j \in \mathfrak{k}[2e_j]$, for $j = 1, \dots, r$ (cf. (3)), one has

$$\begin{aligned} d^c f(\widetilde{K^j}_z) &= -df(I_0 a_* \sinh(2a_j) P^j) = -df(a_* \sinh(2a_j) A_j) = \\ &= -\frac{d}{ds}\Big|_{s=0} f(\exp(H + \sinh(2a_j) s A_j) K) = -\sinh(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(H). \end{aligned} \quad (17)$$

Next we prove statement (i). Let α and γ be distinct roots with $\alpha \in \Sigma^+$ and $\gamma \in \{0\} \cup (\Sigma^+ \setminus \{2e_1, \dots, 2e_r\})$, with the convention $\mathfrak{p}[0] := \mathfrak{a}$. Let $P \in \mathfrak{p}[\alpha]$ and $Q \in \mathfrak{p}[\gamma]$. Write $P = X^\alpha - \theta X^\alpha$, with $X^\alpha \in \mathfrak{g}^\alpha$, and $I_0 Q = Y^\beta - \theta Y^\beta$, with $Y^\beta \in \mathfrak{g}^\beta$, for some $\beta \in \Sigma^+$. Then $a_* P = -\frac{1}{\sinh \alpha(H)} \tilde{K}_z$, with $K = X^\alpha + \theta X^\alpha \in \mathfrak{k}[\alpha]$, and $a_* I_0 Q = -\frac{1}{\sinh \beta(H)} \tilde{C}_k$, with $C = Y^\beta + \theta Y^\beta \in \mathfrak{k}[\beta]$. Therefore

$$\begin{aligned} h_f(a_* P, a_* Q) &= -dd^c f(a_* P, a_* I_0 Q) = \frac{1}{\sinh \alpha(H) \sinh \beta(H)} \frac{d}{dt}\Big|_{t=0} \mu^K(\exp tC \cdot z) = \\ &= \frac{1}{\sinh \alpha(H) \sinh \beta(H)} \frac{d}{dt}\Big|_{t=0} d^c f(\tilde{K}_{\exp tC \cdot z}) \end{aligned}$$

which, by (14), becomes

$$\begin{aligned} &\frac{1}{\sinh \alpha(H) \sinh \beta(H)} \frac{d}{dt}\Big|_{t=0} d^c f(\widetilde{\text{Ad}_{\exp -tC} K}_z) = \\ &= \frac{1}{\sinh \alpha(H) \sinh \beta(H)} \frac{d}{dt}\Big|_{t=0} d^c f(\tilde{K}_z - t[\widetilde{C, K}]_z + o(t^2)) = \\ &= -\frac{1}{\sinh \alpha(H) \sinh \beta(H)} d^c f([\widetilde{C, K}]_z). \end{aligned}$$

Hence

$$h_f(a_* P, a_* Q) = -\frac{1}{\sinh \alpha(H) \sinh \beta(H)} d^c f([\widetilde{C, K}]_z). \quad (18)$$

The brackets

$$[C, K] = ([Y^\beta, X^\alpha] + \theta[Y^\beta, X^\alpha]) + ([Y^\beta, \theta X^\alpha] + \theta[Y^\beta, \theta X^\alpha]),$$

lie in $\mathfrak{k}[\alpha + \beta] + \mathfrak{k}[\alpha - \beta]$. Since $\alpha \in \Sigma^+$ and $\gamma \in \{0\} \cup (\Sigma^+ \setminus \{2e_1, \dots, 2e_r\})$ are distinct, the spaces $\mathfrak{k}[\alpha + \beta]$ and $\mathfrak{k}[\alpha - \beta]$ have zero intersection with $\bigoplus_j \mathfrak{k}[2e_j]$. Then the expression (18) vanishes by (15) and (16), and the spaces $a_* \mathfrak{p}[\alpha]$ and $a_* \mathfrak{p}[\gamma]$ are h_f -orthogonal, as claimed. By the I_0 invariance of h_f , this also implies that $a_* \mathfrak{a}$ is h_f -orthogonal to $\bigoplus_{\alpha \in \Sigma^+} a_* \mathfrak{p}[\alpha]$. This concludes the proof of (i).

Next we examine the Hermitian tensor h_f on the various blocks.

(ii) The Hermitian form h_f on $a_* \mathfrak{a}$.

Let $A_j, A_l \in \mathfrak{a}$. Since $I_0 A_l = -P^l$, one has

$$\begin{aligned} h_f(a_* A_j, a_* A_l) &= -dd^c f(a_* P^l, a_* A_j) = \frac{1}{\sinh(2a_l)} dd^c f((\tilde{K}^l)_z, (\tilde{A}_j)_z) = \\ &= -\frac{1}{\sinh(2a_l)} \frac{d}{dt}\Big|_{t=0} \mu^{K^l}(\exp tA_j \cdot z) = -\frac{1}{\sinh(2a_l)} \frac{d}{dt}\Big|_{t=0} d^c f(\tilde{K}_{\exp tA_j}^l) \\ &= \frac{1}{\sinh(2a_j)} \frac{d}{dt}\Big|_{t=0} df(I_0(\exp(H + tA_j))_* \sinh 2e_j(H + tA_j)) \tilde{P}^l = \\ &= \frac{1}{\sinh(2a_l)} \frac{d}{dt}\Big|_{t=0} \sinh 2e_j(H + tA_j) \frac{\partial \tilde{f}}{\partial a_l}(H + tA_j) = \\ &= \frac{1}{\sinh(2a_l)} (2 \cosh(2a_l) \frac{\partial \tilde{f}}{\partial a_l}(H) \delta_{j,l} + \sinh(2a_l) \frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l}(H)). \end{aligned}$$

The above expression is well defined also for those $H = \sum_j a_j A_j$ with some zero coordinate. Assume for example $a_l = 0$. By the W -invariance, $\tilde{f}$ is even

with respect to the coordinate a_l and consequently its derivative $\frac{\partial \tilde{f}}{\partial a_l}$ vanishes for $a_l = 0$. Therefore

$$\lim_{a_l \rightarrow 0} = 2 \coth(2a_l) \frac{\partial \tilde{f}}{\partial a_l} = \frac{\partial^2 \tilde{f}}{\partial a_l^2},$$

and the above quantity smoothly extends to the hyperplane $a_l = 0$, for all $l = 1, \dots, r$. This concludes the proof of (ii).

(iii) The Hermitian form h_f on $a_*\mathfrak{p}[e_j + e_l]$.

Let $P, Q \in \mathfrak{p}[e_j + e_l]$ be elements of the basis of Remark 2.3 (b), arising from roots $\mu, \nu \in \Delta^+$, respectively, with $\nu \neq \mu, \bar{\mu}$. Then $h_f(a_*P, a_*Q) = 0$, due the fact that for such roots $[Z^\mu \pm \bar{Z}^\mu, Z^\nu \pm \bar{Z}^\nu] = 0$, for all $Z^\mu \in \mathfrak{g}^\mu$ and $Z^\nu \in \mathfrak{g}^\nu$.

Next, let $P, P' \in \mathfrak{p}[e_j + e_l]$ be elements of the basis given in Remark 2.3 (b), arising from the same root $\mu \in \Delta^+$. Write $P = X - \theta X$, with $X = Z^\mu + \bar{Z}^\mu$, and $I_0P = Y - \theta Y$, with $Y = [Z_0, X]$. Likewise write $P' = X' - \theta X'$, with $X' = iZ^\mu - i\bar{Z}^\mu$, and $I_0P' = Y' - \theta Y'$, with $Y' = [Z_0, X']$.

From (18) it follows that

$$h_f(a_*P, a_*P) = -\frac{1}{\sinh(a_j + a_l) \sinh(a_j - a_l)} (d^c f(\widetilde{[C, K]_z})),$$

where $K = X + \theta X$ and $C = Y + \theta Y$. By Lemma 2.4(a)(b), the above expression equals

$$\begin{aligned} & -\frac{1}{\sinh(a_j + a_l) \sinh(a_j - a_l)} (rd^c f(\widetilde{K^j_z}) - sd^c f(\widetilde{K^l_z})) \\ &= \frac{1}{\sinh(a_j + a_l) \sinh(a_j - a_l)} (r \sinh(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(H) - s \sinh(2a_l) \frac{\partial \tilde{f}}{\partial a_l}(H)), \quad r, s \in \mathbb{R}. \end{aligned} \tag{19}$$

In a similar way, one obtains

$$h_f(a_*P', a_*P') = h_f(a_*P, a_*P), \tag{20}$$

and, from Lemma 2.4(c),

$$h_f(a_*P, a_*P') = 0.$$

An argument similar to the one used in (ii) shows that the above expressions (19) and (20) smoothly extend to the hyperplane $(e_j - e_l)(H) = 0$.

Moreover the quantities (19) and (20) are strictly positive for the strictly plurisubharmonic potential ρ of the Killing metric of G/K given in Proposition 5.1. Then (iii) in Lemma 2.5 implies that $r = s > 0$. Finally, as $h_\rho(a_*P, a_*P) = B(P, P)$, a simple computation shows that $r = B(P, P)/b$. This concludes the proof of (iii).

(iv) The Hermitian form h_f on $a_*\mathfrak{p}[e_j]$.

Let $P, Q \in \mathfrak{p}[e_j]$ be elements of the basis given in Remark 2.3 (c), arising from roots $\mu, \nu \in \Delta^+$, respectively, with $\nu \neq \mu, \bar{\mu}$. Then $h_f(a_*P, a_*Q) = 0$, due the fact that for such roots $[Z^\mu \pm \bar{Z}^\mu, Z^\nu \pm \bar{Z}^\nu] = 0$, for all $Z^\mu \in \mathfrak{g}^\mu$ and $Z^\nu \in \mathfrak{g}^\nu$. In addition, by the I_0 -invariance of h_f one has $h_f(a_*P, a_*I_0P) = 0$.

In order to compute $h_f(a_*P, a_*P)$, write $P = X - \theta X$ and $I_0P = Y - \theta Y$, with $X = Z^\mu + \bar{Z}^\mu$ and $Y = iZ^\mu - i\bar{Z}^\mu$ (Lemma 2.2 (c)). Then, from (18) it follows that

$$h_f(a_*P, a_*P) = -\frac{1}{\sinh^2(a_j)} d^c f(\widetilde{[C, K]_z}),$$

for $K = X + \theta X$ and $C = Y + \theta Y$. By Lemma 2.4 (d)(e), one obtains

$$h_f(a_*P, a_*P) = -\frac{1}{\sinh^2(a_j)} d^c f(t\tilde{K}^j_z) = 2t \coth(a_j) \frac{\partial \tilde{f}}{\partial a_j}(H), \quad t \in \mathbb{R}.$$

Note that the above expression smoothly extends on $\mathcal{D}_a$, since $\frac{\partial \tilde{f}}{\partial a_j} = 0$ on the hyperplane $a_j = 0$. Moreover, by (i) of Lemma 2.5, for $a_j > 0$ one has $\frac{\partial \tilde{f}}{\partial a_j} > 0$.

As in the previous case, one shows that $t = B(P, P)/b$ by computing the above quantity for the strictly plurisubharmonic potential ρ of the Killing metric of G/K given in Proposition 5.1. This completes the proof of statement (iv). $\square$

Remark. The Levi form $L_f^\mathbb{C}$ of f is given by

$$L_f^\mathbb{C}(Z, \bar{W}) = 2(h_f(X, Y) + ih_f(X, I_0Y)),$$

where $Z = X - I_0X$ and $W = Y - I_0Y$ are elements in $(\mathfrak{p}^\mathbb{C})^{1,0}$. One easily sees that $L_f^\mathbb{C}$ is (strictly) positive definite if and only if h_f is (strictly) positive definite.

4. K -INVARIANT PSH FUNCTIONS VS. W -INVARIANT LOGCVX FUNCTIONS

Let G/K be an irreducible non-compact Hermitian symmetric space of rank r and let $D \subset G/K$ be a Stein, K -invariant domain. The goal of this section is to prove a characterization of various classes of K -invariant plurisubharmonic functions on D in terms of appropriate conditions of the corresponding functions on $\mathcal{D}_a$ (see (10)). As an application, we reproduce the characterization of Stein K -invariant domains in G/K (Cor. 4.8), outlined in [BeDa91], Thm. 3' and Thm. 4 (see also [FeHu93]).

In the smooth case we prove that a smooth K -invariant function f of D is strictly plurisubharmonic if and only if the associated function $\tilde{f}$ (see (11)) satisfies a positivity condition arising from Proposition 3.1.

Denote by Δ^r the orbit of the base point $eK \in G/K$ under the commuting $SL_2(\mathbb{R})$'s generated by the triples defined in (2)-(3). It is well-known (cf. [Wol72]) that Δ^r is biholomorphic to the unit polydisk in $\mathbb{C}^r$. One has $\Delta^r = T \exp \mathfrak{a}K$, where $T \cong (S^1)^r$ is the r -dimensional torus in K whose Lie algebra is generated by $K^1, \dots, K^r$, and

$$\exp(a_1, \dots, a_r)K = (\tanh(a_1), \dots, \tanh(a_r)), \quad \text{for } (a_1, \dots, a_r) \in \mathfrak{a}.$$

The polydisk Δ^r is a “thick slice” for the K -action in G/K , in the sense that $K \cdot \Delta^r = G/K$. If D is a K -invariant domain in G/K , then the Reinhardt domain associated to D is defined as

$$R := D \cap \Delta^r \quad \text{and} \quad D = K \cdot R.$$

We will show that if D is Stein, then R is necessarily connected. (It should be remarked that, despite its appellation, a Reinhardt “domain” is open in $\mathbb{C}^r$ but need not be connected).

For a Reinhardt domain R in Δ^r , define the set $\mathcal{D} = \{(a_1, \dots, a_r) \in \mathbb{R}^r : (\tanh a_1, \dots, \tanh a_r) \in R\}$, with the property that the image of the map

$$\mathcal{D} \rightarrow R \quad (a_1, \dots, a_r) \rightarrow (\tanh a_1, \dots, \tanh a_r)$$

coincides with $R \cap \mathbb{R}^r$. One has $R = T \cdot (R \cap \mathbb{R}^r)$, with $T \cong (S^1)^r$. Given a smooth T -invariant function f on R define $\tilde{f} : \mathcal{D} \rightarrow \mathbb{R}$ by

$$\tilde{f}(a_1, \dots, a_r) = f(\tanh a_1, \dots, \tanh a_r).$$

By the T -invariance of f , the function $\tilde{f}$ is $(\mathbb{Z}^2)^r$ -invariant.

Denote by $\text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$ the class of smooth functions on $\mathcal{D}$ which are even in each variable and such that the form defined in (ii) of Proposition 3.1 is strictly positive definite. The next proposition characterizes T -invariant smooth strictly plurisubharmonic functions on R by elements in $\text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$. It is an intermediate step in the proof of the main theorem in the smooth case, but it may be of independent interest in the context of Reinhardt domains.

Proposition 4.1. *Let f be a smooth T -invariant function on a Reinhardt domain R in Δ^r . Then f is strictly plurisubharmonic if and only if $\tilde{f}$ belongs to $\text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$.*

Proof. In polar coordinates (ρ_j, θ_j) , with $z_j = \rho_j e^{i\theta_j} \neq 0$, one has

$$\partial_{z_j} = \frac{e^{-i\theta_j}}{2\rho_j}(\rho_j \partial_{\rho_j} - i\partial_{\theta_j}) \quad \partial_{\bar{z}_j} = \frac{e^{i\theta_j}}{2\rho_j}(\rho_j \partial_{\rho_j} + i\partial_{\theta_j}).$$

One easily sees that, for $z_j z_l \neq 0$,

$$4 \frac{\partial^2 f}{\partial \bar{z}_j \partial z_l}(z_1, \dots, z_r) = \frac{1}{\rho_j} \frac{\partial f}{\partial \rho_j}(\rho_1, \dots, \rho_r) \delta_{jl} + e^{i(\theta_j - \theta_l)} \frac{\partial^2 f}{\partial \rho_j \partial \rho_l}(\rho_1, \dots, \rho_r). \quad (21)$$

The above quantity extends smoothly through the hyperplanes $z_j = 0$ (and therefore to the whole domain) whenever $j = l$, while

$$4 \frac{\partial^2 f}{\partial \bar{z}_j \partial z_l}(z_1, \dots, z_r) = 0, \quad \text{for } j \neq l \text{ and } z_j z_l = 0.$$

For $\rho_1 = \tanh a_1, \dots, \rho_r = \tanh a_r$, one has

$$\frac{\partial \tilde{f}}{\partial a_j}(a_1, \dots, a_r) = \frac{\partial f}{\partial \rho_j}(\tanh a_1, \dots, \tanh a_r) \frac{1}{\cosh^2 a_j} \quad (22)$$

$$\frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l}(H) = \frac{\partial^2 f}{\partial \rho_j \partial \rho_l}(\tanh a_1, \dots, \tanh a_r) \frac{1}{\cosh^2 a_j \cosh^2 a_l} - \delta_{jl} \frac{\partial f}{\partial \rho_j}(\tanh a_1, \dots, \tanh a_r) \frac{2 \sinh a_j}{\cosh^3 a_j}, \quad (23)$$

and likewise

$$\frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l}(H) = 0, \quad \text{for } j \neq l \text{ and } a_j a_l = 0.$$

A simple computation combining formulas (22) and (23) with (21), shows that $4 \frac{\partial^2 f}{\partial \bar{z}_j \partial z_l}(z_1, \dots, z_r)$ is given by

$$\begin{cases} \cosh^4 a_j (2 \coth(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(a_1, \dots, a_r) + \frac{\partial^2 \tilde{f}}{\partial a_j^2}(a_1, \dots, a_r)), & \text{for } j = l, \\ \cosh^2 a_j e^{i\theta_j} \cosh^2 a_l e^{-i\theta_l} \frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l}(a_1, \dots, a_r), & \text{for } j \neq l \text{ and } z_j z_l \neq 0, \\ 0, & \text{for } j \neq l \text{ and } z_j z_l = 0. \end{cases}$$

Then, for $(z_1, \dots, z_r) \in R$, one has

$$\left(4 \frac{\partial^2 f}{\partial \bar{z}_j \partial z_l}\right)_{j,l} = C \left(\frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l} + \delta_{jl} 2 \coth(2a_j) \frac{\partial \tilde{f}}{\partial a_j} \right)_{j,l} \bar{C},$$

where C is the diagonal matrix with diagonal entries

$$c_{jj} = \begin{cases} \cosh^2(a_j) e^{i\theta_j}, & \text{for } z_j \neq 0, \\ \cosh^2(a_j), & \text{for } z_j = 0. \end{cases}$$

It follows that f is strictly plurisubharmonic if and only if $\tilde{f}$ belongs to the class $\text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$. $\square$

Let R be a Reinhardt domain in $(\Delta^*)^r$ and let

$$\mathcal{D}_{\log} := \{(s_1, \dots, s_r) \in (\mathbb{R}^{<0})^r : (e^{s_1}, \dots, e^{s_r}) \in R\}$$

be its logarithmic image. For a T -invariant function f on R , define $\hat{f}: \mathcal{D}_{\log} \rightarrow \mathbb{R}$ by

$$\hat{f}(s_1, \dots, s_r) := f(e^{s_1}, \dots, e^{s_r}). \quad (24)$$

It is well known that if f is smooth, then it is strictly plurisubharmonic if and only if $\hat{f}$ has strictly positive definite Hessian. The next remarks elucidate the significance of the class $\text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$.

Remark 4.2. *Let R be a Reinhardt domain in $(\Delta^*)^r$ and let f be a smooth T -invariant function on R . Then $\tilde{f}$ belongs to $\text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$ if and only if the smooth function $\hat{f}$ has everywhere strictly positive Hessian.*

Proof. One has

$$\frac{\partial \hat{f}}{\partial s_j}(s_1, \dots, s_r) = \frac{\partial f}{\partial \rho_j}(e^{s_1}, \dots, e^{s_r}) e^{s_j}, \quad (25)$$

$$\frac{\partial^2 \hat{f}}{\partial s_j \partial s_l}(s_1, \dots, s_r) = \frac{\partial^2 f}{\partial \rho_j \partial \rho_l}(e^{s_1}, \dots, e^{s_r}) e^{s_j} e^{s_l} + \delta_{jl} \frac{\partial f}{\partial \rho_j}(e^{s_1}, \dots, e^{s_r}) e^{s_j}. \quad (26)$$

Then, by letting $e^{s_1} = \tanh a_1, \dots, e^{s_r} = \tanh a_r$, with $a_1, \dots, a_r > 0$, and combining formulas (25) and (26) with (22) and (23), one obtains

$$2 \coth(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(H) \delta_{jl} + \frac{\partial^2 \tilde{f}}{\partial a_j \partial a_l}(H) = \frac{4}{\sinh 2a_j \sinh 2a_l} \frac{\partial^2 \hat{f}}{\partial s_j \partial s_l}(s_1, \dots, s_r).$$

Hence $\tilde{f} \in \text{LogConv}^{\infty,+}(\mathcal{D})^{(\mathbb{Z}_2)^r}$ if and only if $\hat{f}$ has everywhere strictly positive Hessian. $\square$

Remark 4.3. Let R be an arbitrary Reinhardt domain and let $f|_{R \cap (\Delta^*)^r}$ denote the restriction of f to $R \cap (\Delta^*)^r$. The strict positivity of the Hessian of $f|_{R \cap (\Delta^*)^r}$ on $R \cap (\Delta^*)^r$ does not imply the strict plurisubharmonicity of f on the coordinate hyperplanes (and therefore on the whole R). For instance, despite the fact that it has strictly positive Hessian on $R \cap (\Delta^*)^r$, the function $g(z) = |z|^4$ is not plurisubharmonic at $z = 0$. In contrast, this fact is detected by the vanishing of the form

$$\frac{\partial^2 \tilde{g}}{\partial a \partial a} + 2 \coth(2a) \frac{\partial \tilde{g}}{\partial a} = 16 \frac{\tanh^2 a}{\cosh^4 a},$$

at $a = 0$, which shows that the associated function $\tilde{g}(a) = \tanh(a)^4$ does not belong to $\text{LogConv}^{\infty,+}(\mathbb{R})^{(\mathbb{Z}_2)}$.

Let $R \subset \Delta^r$ be a Reinhardt domain associated to a K -invariant domain in G/K . In this case, R is also invariant under coordinate permutations, which arise from the Weyl group action on $\mathfrak{a}$. If such a Reinhardt domain is Stein, then there are two possibilities:

- (a) R intersects the coordinate hyperplanes. Then it is complete (cf. [Car73], Thm. 2.12). In particular it contains the origin and is connected.
- (b) R does not intersect the coordinate hyperplanes, i.e. $R \subset (\Delta^*)^r$. Then R is logarithmically convex.

The next proposition shows that a Stein Reinhardt domain R associated to a Stein K -invariant domain $D \subset G/K$ is necessarily connected (even when $0 \notin R$), a fact already pointed out in [BeDa91], Thm. 3'.

Proposition 4.4. Let D and R be as above and let $f : D \rightarrow \mathbb{R}$ be a smooth, K -invariant strictly plurisubharmonic exhaustion of D .

- (i) If R contains the origin, then R is connected and $\tilde{f}$ has a unique minimum point at the origin of $\mathcal{D}_{\mathfrak{a}}$.
- (ii) If R does not contain the origin, then $\tilde{f}$ has a unique minimum point on the diagonal line $\{a_1 = \dots = a_r\}$ of $\mathcal{D}_{\mathfrak{a}}$. In particular R is connected. In this case G/K is necessarily of tube type.

Proof. The minimum set of a K -invariant exhaustion function f of D intersects R in a non-empty T -invariant set. Since $R = T \cdot \exp \mathcal{D}_{\mathfrak{a}} K$, a point $H \in \mathcal{D}_{\mathfrak{a}}$ is a minimum of $\tilde{f}$ if and only if $\exp(H)K \in R$ is a minimum of $f|_R$, the restriction of f to R .

(i) We already observed that R is connected. Assume that $\tilde{f}$ has a minimum point $H = (a_1, \dots, a_r)$, different from the origin. Then the restriction $f|_R$ of f to the Reinhardt domain R has a minimum point in $P = \exp(H)K$. For ε small enough there is a holomorphic immersion

$$\iota : \Delta_{1+\varepsilon} \rightarrow R, \quad z \rightarrow zP$$

from the disc of radius $1 + \varepsilon$ to R . The pull-back $f \circ \iota$ of f via ι is a smooth strictly subharmonic S^1 -invariant function. Hence it has a minimum point

in 0 and, by construction, in 1. It follows that $f \circ \iota$ is necessarily constant, contradicting the fact that it is strictly subharmonic.

(ii) Let $H = (a_1, \dots, a_r)$ be a minimum point of $\tilde{f}$. In this case, all a_j 's are different from 0. As a consequence $2 \coth(a_j) \frac{\partial \tilde{f}}{\partial a_j}(H) = 0$, for $j = 1, \dots, r$. By (iv) of Proposition 3.1, in the non-tube case this contradicts the strict plurisubharmonicity of f , implying that the space G/K is necessarily of tube type. The strict plurisubharmonicity of f along with (iii) of Proposition 3.1, also implies that $a_j = a_k$ for every $j, k = 1, \dots, r$. Hence H lies on the diagonal of $\mathfrak{a}$. The uniqueness of the minimum follows from standard arguments as in [AzLo93] or by the the following direct argument.

Recall that $\mathcal{D}_{\log}$ is convex by the Steinness of D . By Remark 4.2, the associated function $\hat{f}$ has everywhere strictly positive definite Hessian. In particular its restriction to the diagonal $\mathcal{D}_{\log} \cap \{s_1 = \dots = s_r\}$ is a strictly convex exhaustion function. Hence it has a unique minimum, implying that $\tilde{f}$ has a unique minimum on $\mathcal{D}_{\mathfrak{a}} \cap \{a_1 = \dots = a_r\}$. $\square$

Consider the following classes of functions:

- $C^0(\mathcal{D}_{\mathfrak{a}})^W$: continuous W -invariant functions on $\mathcal{D}_{\mathfrak{a}}$,
- $C^\infty(\mathcal{D}_{\mathfrak{a}})^W$: smooth W -invariant functions on $\mathcal{D}_{\mathfrak{a}}$,
- $C^0(D)^K$: continuous K -invariant functions on D ,
- $C^\infty(D)^K$: smooth K -invariant functions on D .

Since the K -action on D is proper and every K -orbit intersects the slice $\exp \mathcal{D}_{\mathfrak{a}} K$ in a W -orbit, it is easy to check that the map $f \rightarrow \tilde{f}$ is a bijection from $C^0(D)^K$ onto $C^0(\mathcal{D}_{\mathfrak{a}})^W$. By Theorem 4.1 in [Fle78] (see also [Dad82]) such a map is also a bijection from $C^\infty(D)^K$ onto $C^\infty(\mathcal{D}_{\mathfrak{a}})^W$. Define

- $\text{LogConv}^{\infty,+}(\mathcal{D}_{\mathfrak{a}})^W$: smooth, W -invariant functions on $\mathcal{D}_{\mathfrak{a}}$ such that the form defined in (ii) of Proposition 3.1 is strictly positive definite,
- $P^{\infty,+}(D)^K$: smooth, K -invariant, strictly plurisubharmonic functions (i.e. with strictly positive definite Levi form) on D .

Our first result is the following theorem.

Theorem 4.5. *Let D be a Stein K -invariant domain in an irreducible non-compact Hermitian symmetric space G/K of rank r . Then $f \in P^{\infty,+}(D)^K$ if and only if $\tilde{f} \in \text{LogConv}^{\infty,+}(\mathcal{D}_{\mathfrak{a}})^W$.*

Proof. By (ii) of Proposition 3.1, if f is strictly plurisubharmonic on D , then $\tilde{f} \in \text{LogConv}^{\infty,+}(\mathcal{D}_{\mathfrak{a}})^W$.

Conversely, assume that $\tilde{f} \in \text{LogConv}^{\infty,+}(\mathcal{D}_{\mathfrak{a}})^W$ and $r > 1$. We need to show that the terms in (iii) and (iv) of Proposition 3.1 are strictly positive, the ones in (iv) occurring only in the non-tube case.

For the terms in (iii), without loss of generality, it is sufficient to consider the case $r = 2$, and $H = (a_1, a_2) \in \mathfrak{a}^+$, where $a_1 \geq a_2 \geq 0$. Assume first $a_1 > a_2 > 0$. Then $(\tanh a_1, \tanh a_2) = (e^{s_1}, e^{s_2}) \in R^*$, where R is the Reinhardt

domain associated to D . Let $d_0 < 0$ and $t_0 > 0$ be real numbers defined by $(s_1, s_2) = (d_0 + t_0, d_0 - t_0)$.

Denote by SSC (smooth stably convex) those smooth functions with everywhere positive definite Hessian. The function $\hat{f}$, which is invariant under coordinate permutations, is SSC by Remark 4.2. Therefore $g(t) := \hat{f}(d_0 + t, d_0 - t)$ is even and SSC. Consequently, for $t_0 > 0$, the inequality

$$g'(t_0) = \frac{\partial f}{\partial \rho_1}(e^{d_0+t_0}, e^{d_0-t_0})e^{d_0+t_0} - \frac{\partial f}{\partial \rho_2}(e^{d_0+t_0}, e^{d_0-t_0})e^{d_0-t_0} > 0,$$

holds true, which combined with formulas (22) yields the desired result.

$$g'(t_0) = \frac{1}{2}(\sinh(2a_j) \frac{\partial \tilde{f}}{\partial a_j}(H) - \sinh(2a_l) \frac{\partial \tilde{f}}{\partial a_l}(H)) > 0.$$

Next we need to estimate the terms (iii) of Proposition 3.1, when H lies on the boundary of the Weyl chamber $\mathfrak{a}^+$. Consider $H = (a, a)$, with $a \neq 0$. Set $\tanh a = e^{d_0}$ and $(\tanh a_1, \tanh a_2) = (e^{d_0+t}, e^{d_0-t})$, and recall that $g'(0) = 0$. Then the corresponding term in (iii) of Proposition 3.1 is the limit

$$\begin{aligned} \frac{1}{2} \lim_{t \rightarrow 0} \frac{g'(t)}{\sinh(a_1 + a_2) \sinh(a_1 - a_2)} &= \frac{1}{4} \lim_{t \rightarrow 0} \frac{g'(t)}{t} \frac{s_1 - s_2}{\sinh(a_1 + a_2) \sinh(a_1 - a_2)} \\ &= \frac{1}{4} \frac{g''(0)}{\sinh(2a) \cosh(2a)} \lim_{a_1 - a_2 \rightarrow 0} \frac{\log \tanh a_1 - \log \tanh a_2}{a_1 - a_2} = g''(0)c(a), \end{aligned}$$

which is positive since $c(a)$ is a positive real number and $g''(0) > 0$ (g is even and SSC).

If $H = (a_1, 0) \in \mathcal{D}_a$, with $a_1 > 0$, then the Reinhardt domain R associated to D is necessarily complete and the term to be evaluated reduces to

$$\frac{1}{\sinh^2 a_1} \sinh(2a_1) \frac{\partial \tilde{f}}{\partial a_1}(a_1, 0).$$

Moreover

$$2 \coth(2a_1) \frac{\partial \tilde{f}}{\partial a_1}(a_1, 0) + \frac{\partial^2 \tilde{f}}{\partial a_1^2}(a_1, 0) > 0,$$

implying that the function $s_1 \rightarrow f(e^{s_1}, 0)$ is SSC (cf. Rem. 4.2). Since R is complete, then $\lim_{s_1 \rightarrow -\infty} f(e^{s_1}, 0)$ is finite. As a consequence $s_1 \rightarrow f(e^{s_1}, 0)$ is strictly increasing and so is $a_1 \rightarrow \tilde{f}(a_1, 0) = f(\tanh a_1, 0)$. Hence $\frac{\partial \tilde{f}}{\partial a_1}(a_1, 0)$ is positive, as wished.

Finally note that for $a_1 = a_2 = 0$ the analytic extension of our term is given by

$$2 \frac{\partial^2 \tilde{f}}{\partial a_1^2}(0, 0) = 2 \frac{\partial^2 \tilde{f}}{\partial a_2^2}(0, 0),$$

which is strictly positive by assumption.

We are left to examine the terms in (iv), which only appear in the non-tube case. Our arguments are similar to the ones used above. By Proposition 4.4, the Reinhardt domain R associated to D is complete. Then $\lim_{s_j \rightarrow -\infty} \hat{f}(s_1, \dots, s_j, \dots, s_r)$ is finite. Since $\hat{f}$ is SSC, the function $s_j \rightarrow \hat{f}(s_1, \dots, s_j, \dots, s_r)$ is strictly increasing and so is $a_j \rightarrow \tilde{f}(a_1, \dots, a_j, \dots, a_r)$. Hence

$$2 \coth(a_j) \frac{\partial \tilde{f}}{\partial a_j}(a_1, \dots, a_r) > 0, \quad \text{for } a_j > 0.$$

The limit

$$\lim_{a_j \rightarrow 0} 2 \coth(a_j) \frac{\partial \tilde{f}}{\partial a_j}(a_1, \dots, a_j, \dots, a_r) = 2 \frac{\partial^2 \tilde{f}}{\partial a_j^2}(a_1, \dots, 0, \dots, a_r)$$

is strictly positive as well, by assumption. $\square$

Consider the $(T \ltimes \mathcal{S}_r)$ -action on Δ^r , where $\mathcal{S}_r$ denotes the group of coordinate permutations. As a consequence of Proposition 4.1 one has the following corollary.

Corollary 4.6. *Let D be a Stein K -invariant domain in an irreducible non-compact Hermitian symmetric space G/K and let R be the associated Reinhardt domain. The map $f \rightarrow f|_R$ is a bijection between $P^{\infty,+}(D)^K$ and $P^{\infty,+}(R)^{T \ltimes \mathcal{S}_r}$.*

Remark 4.7. *If R does not contain the origin, then, by Remark 4.2, the condition $f \in P^{\infty,+}(D)^K$ is also equivalent to requiring that the smooth invariant function $\hat{f}$ has strictly positive definite Hessian on $\mathcal{D}_{\log}$.*

Corollary 4.8. *(see [BeDa91], Thm.3' and Thm.4) Let D be a Stein K -invariant domain in an irreducible non-compact Hermitian symmetric space G/K and let R be the associated Reinhardt domain. Then*

- (i) *If G/K is of tube type, then D is Stein if and only if R is Stein and connected.*
- (ii) *If G/K is not of tube type, then D is Stein if and only if R is Stein and complete. In particular R contains the origin and is connected.*

Proof. By Proposition 4.4, if D is Stein then the intersection $R = D \cap \Delta^r$ is Stein, connected and, in the non-tube case, complete. Conversely, let R be a Stein, connected Reinhardt domain, invariant under coordinate permutations which, in the non-tube case, is also assumed to be complete. Let f be a smooth, strictly plurisubharmonic exhaustion function f of R . By averaging, f may be assumed to be invariant with respect to T and to coordinate permutations. Proposition 4.1 implies that the function $\hat{f}: \mathcal{D}_a \rightarrow \mathbb{R}$, associated to f , belongs to $\text{LogConv}^{\infty,+}(\mathcal{D}_a)^W$. By Theorem 4.5, $\hat{f}$ extends to a smooth, K -invariant, strictly plurisubharmonic exhaustion function of D . Hence D is Stein. $\square$

Remark 4.9. The envelope of holomorphy of a K -invariant domain D in G/K is described, without proof, in terms of the associate Reinhardt domain R in Theorem 5 of [BeDa91]:

if G/K is of tube type, then $\hat{D} = K \cdot \tilde{R}$, where $\tilde{R}$ is the smallest connected Stein, Reinhardt domain containing R ;

if G/K is not of tube type, then $\hat{D} = K \cdot \tilde{R}$, where $\tilde{R}$ is the smallest connected and complete Stein, Reinhardt domain containing R .

One can easily prove the above theorem in the following cases. If R is connected and intersects the coordinate hyperplanes, then the envelope of holomorphy $\hat{D}$ of D is schlicht and coincides with $K \cdot \hat{R}$, where $\hat{R}$ is the envelope of holomorphy of R . Indeed, by [Car73], Thm.2.12. the envelope of holomorphy $\hat{R}$ of R , is

schlicht and is a Stein, complete Reinhardt domain in Δ^r . As a consequence, the invariant domain $K \cdot \hat{R}$ is contained in $\hat{D}$, and it is Stein by Corollary 4.8. It follows that $\hat{D} = K \cdot \hat{R}$.

By the same argument, this equality holds true also in the case when G/K is of tube type and D is any K -invariant domain such that R is connected.

Our next goal is to extend the characterization of smooth, K -invariant, strictly plurisubharmonic functions on D obtained in Theorem 4.5 to some wider classes of K -invariant functions. Namely:

- $P(D)^K$: plurisubharmonic, K -invariant functions on D ,
- $P^\infty(D)^K$: smooth, plurisubharmonic, K -invariant functions on D ,
- $P^+(D)^K$: functions which, on every relatively compact K -invariant domain C in D , are the sum $g + h$, for some $g \in P(C)^K$ and $h \in P^{\infty,+}(C)^K$.

In order to do that we need to define the appropriate classes of functions on the associated domain $\mathcal{D}_a$:

- $\text{LogConv}(\mathcal{D}_a, [-\infty, \infty))^W$: limits of decreasing sequences in $\text{LogConv}^{\infty,+}(\mathcal{D}_a)^W$,
- $\text{LogConv}^\infty(\mathcal{D}_a)^W$: smooth functions in $\text{LogConv}(\mathcal{D}_a, [-\infty, \infty))^W$,
- $\text{LogConv}^+(\mathcal{D}_a, [-\infty, \infty))^W$: functions which, on every relatively compact W -invariant domain $\mathcal{C}$ of $\mathcal{D}_a$, are the sum $\tilde{g} + \tilde{h}$ for some $\tilde{g} \in \text{LogConv}(\mathcal{C}, [-\infty, \infty))^W$ and $\tilde{h} \in \text{LogConv}^{\infty,+}(\mathcal{C})^W$.

Remark 4.10. (i) The class $\text{LogConv}^\infty(\mathcal{D}_a)^W$ coincides with the family of smooth W -invariant functions on $\mathcal{D}_a$ for which the form in (ii) of Proposition 3.1 is non-negative. One inclusion is clear. Conversely, if $\tilde{f}$ is smooth and the form in (ii) of Proposition 3.1 is non-negative, then $\tilde{f}$ is the limit of the sequence $\tilde{f}_n(a_1, \dots, a_r) = \tilde{f}(a_1, \dots, a_r) + \frac{1}{n} \sum a_j^2$. Hence $\tilde{f}$ belongs to $\text{LogConv}^\infty(\mathcal{D}_a)^W$. In particular

$$\text{LogConv}^{\infty,+}(\mathcal{D}_a)^W \subset \text{LogConv}^\infty(\mathcal{D}_a)^W.$$

(ii) The class $P^+(D)^K$ coincides with the family of functions which are locally the sum of some g plurisubharmonic and h smooth strictly plurisubharmonic, i.e. the strictly plurisubharmonic functions according to [Gun90], Def. 1, Sect. L, p. 118. Indeed, assume that f is strictly plurisubharmonic according to such definition. Choose a K -invariant, smooth strictly plurisubharmonic function ψ on D and let C be a relatively compact K -invariant domain of D . Then there exists $\varepsilon > 0$ such that $f - \varepsilon\psi$ is plurisubharmonic on C . That is, $f = g + \varepsilon\psi$, with g psh and K -invariant on C .

The following lemma shows that all functions in the above classes are continuous.

Lemma 4.11. *Let R be a Reinhardt domain.*

- i) Any T -invariant plurisubharmonic function f on R is continuous. Its pluripolar set is the union of the intersections of R with some coordinate subspaces.
- ii) The class $\text{LogConv}(\mathcal{D}_a, [-\infty, \infty))^W$ is contained in $C^0(\mathcal{D}_a, [-\infty, \infty))^W$.

Proof. (i) First consider the case $r = 1$. On $R^* = R \setminus \{0\}$ one has $f(z) = \hat{f}(\log |z|)$, with $\hat{f}$ convex. Hence the restriction of f to R^* is continuous. If $0 \in R$, by the upper semicontinuity of subharmonic functions, one has $f(0) = \limsup_{z \rightarrow 0} f(z)$. Assume by contradiction that

$$\liminf_{z \rightarrow 0} f(z) < f(0).$$

Then there exists $z_1 \in R$ close to the origin such that $f(z_1) < f(0)$. By the submean value property and the S^1 -invariance of f one has

$$f(0) \leq \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta} z_1) d\theta = f(z_1) < f(0),$$

which is a contradiction.

For $r = 2$, an argument analogous to the above one shows that f is continuous on $R^* = R \cap (\Delta^*)^r$. We now prove continuity on the coordinate hyperplanes $\{(z, w) \in R : zw = 0\}$ (on each hyperplane it can be constant and equal to $-\infty$). Assume by contradiction that there exists $(z_0, 0) \in R$ such that

$$\limsup_{(z,w) \rightarrow (z_0,0)} f(z, w) - \liminf_{(z,w) \rightarrow (z_0,0)} f(z, w) > \varepsilon > 0. \quad (27)$$

Since $\limsup_{(z,w) \rightarrow (z_0,0)} f(z, w) = f(z_0, 0)$ by plurisubharmonicity and f is continuous on the hyperplane $w = 0$, there exists a neighborhood B of z_0 in $\mathbb{C}$ such that

$$\limsup_{(z,w) \rightarrow (z_0,0)} f(z, w) - \varepsilon/2 = f(z_0, 0) - \varepsilon/2 < f(\zeta, 0),$$

for every $\zeta \in B$. By (27), we can choose (ζ_1, w_1) close to $(z_0, 0)$ such that $\zeta_1 \in B$ and

$$f(\zeta_1, w_1) < \liminf_{(z,w) \rightarrow (z_0,0)} f(z, w) + \varepsilon/2 < \limsup_{(z,w) \rightarrow (z_0,0)} f(z, w) - \varepsilon/2.$$

Then, by the submean value property for subharmonic functions and by the T -invariance of f one has

$$\begin{aligned} \limsup_{(z,w) \rightarrow (z_0,0)} f(z, w) - \varepsilon/2 &< f(\zeta_1, 0) \leq \frac{1}{2\pi} \int_0^{2\pi} f(\zeta_1, e^{i\theta} w_1) d\theta = \\ &= f(\zeta_1, w_1) < \limsup_{(z,w) \rightarrow (z_0,0)} f(z, w) - \varepsilon/2. \end{aligned}$$

giving a contradiction.

The above argument also shows that the pluripolar set of f consists of either the origin, or the intersection of R with one of the coordinate lines $\{z = 0\}$, $\{w = 0\}$ or with both of them.

Now we can proceed inductively and obtain the statement for $r > 2$.

(ii) By Theorem 4.5, to a decreasing sequence $\tilde{f}_n$ of functions in $\text{LogConv}^{\infty,+}(\mathcal{D}_a)^W$ there corresponds a decreasing sequence f_n in $P^{\infty,+}(D)^K$, whose limit f necessarily belongs to $P(D)^K$. The restriction $f|_R$ of f to R is a plurisubharmonic

T -invariant function. So (i) implies that $f|_R$ is continuous and consequently so is the corresponding $\tilde{f}$ in $\text{LogConv}(\mathcal{D}_a, [-\infty, \infty))^W$, which is the limit of the $\tilde{f}_n$. $\square$

Summarizing, the following inclusions hold true

$$\begin{array}{ccccc} \text{LogConv}^+(\mathcal{D}_a, [-\infty, \infty))^W & \subset & \text{LogConv}(\mathcal{D}_a, [-\infty, \infty))^W & \subset & C^0(\mathcal{D}_a, [-\infty, \infty))^W \\ \cup & & \cup & & \cup \\ \text{LogConv}^{\infty,+}(\mathcal{D}_a)^W & \subset & \text{LogConv}^\infty(\mathcal{D}_a)^W & \subset & C^\infty(\mathcal{D}_a)^W \end{array}$$

and our complete result is stated in the next theorem.

Theorem 4.12. *Let D be a Stein K -invariant domain in an irreducible non-compact Hermitian symmetric space G/K . The map $f \rightarrow \tilde{f}$ is a bijection between the following classes of functions*

- (i) $P^{\infty,+}(D)^K$ and $\text{LogConv}^{\infty,+}(\mathcal{D}_a)^W$,
- (ii) $P(D)^K$ and $\text{LogConv}(\mathcal{D}_a, [-\infty, \infty))^W$,
- (iii) $P^\infty(D)^K$ and $\text{LogConv}^\infty(\mathcal{D}_a)^W$,
- (iv) $P^+(D)^K$ and $\text{LogConv}^+(\mathcal{D}_a, [-\infty, \infty))^W$.

In particular K -invariant plurisubharmonic functions on D are continuous.

Proof. (i) is the content of Theorem 4.5. By averaging over K , a K -invariant, plurisubharmonic function on D is the decreasing limit of smooth K -invariant, strictly plurisubharmonic functions (cf. [Gun90], Sect. K). Then the assert (ii) follows from (i). As smooth K -invariant functions on D correspond to smooth W -invariant functions on $\mathcal{D}_a$, the previous argument also proves statement (iii). Finally (iv) follows from the definitions of $\text{LogConv}^+(\mathcal{D})^W$ and $P^+(D)^K$, by averaging all the involved functions over K . $\square$

Let $T \ltimes \mathcal{S}_r$ act on Δ^r as in Corollary 4.6. The previous theorem can be reformulated as follows.

Theorem 4.13. *Let D be a Stein K -invariant domain in an irreducible non-compact Hermitian symmetric space G/K and let R be the associated Reinhardt domain. The map $f \rightarrow f|_R$ is a bijection between*

- (i) $P^{\infty,+}(D)^K$ and $P^{\infty,+}(R)^{T \ltimes \mathcal{S}_r}$,
- (ii) $P(D)^K$ and $P(R)^{T \ltimes \mathcal{S}_r}$,
- (iii) $P^\infty(D)^K$ and $P^\infty(R)^{T \ltimes \mathcal{S}_r}$,
- (iv) $P^+(D)^K$ and $P^+(R)^{T \ltimes \mathcal{S}_r}$.

5. APPENDIX: A K -INVARIANT POTENTIAL FOR THE KILLING METRIC: THE LOGARITHM OF THE BERGMAN KERNEL FUNCTION.

Let G/K be an irreducible non-compact Hermitian symmetric space. The Killing form B of $\mathfrak{g}$, restricted to $\mathfrak{p}$, induces a G -invariant Kähler metric h on G/K . In this section we exhibit a K -invariant potential ρ of the Killing metric h in a Lie theoretical fashion. As such a K -invariant potential is unique up to an additive constant (Rem. 5.2), then, up to an additive constant, it coincides with the logarithm of the Bergman kernel function (Cor. 5.3).

In order to define ρ , recall the decomposition $G = K \exp \mathfrak{a} K$ and write an element of G/K as kaK , where $k \in K$ and $a = \exp H$, with $H = \sum_j a_j A_j \in \mathfrak{a}$.

Proposition 5.1. *Let $\hat{\rho}$ be a real valued function satisfying $\hat{\rho}'(t) = \frac{\cosh t - 1}{\sinh t}$. Then*

(i) *the K -invariant function $\rho : G/K \rightarrow \mathbb{R}$ defined by*

$$\rho(kaK) := \frac{1}{4} \sum_{j=1}^r \hat{\rho}(2a_j) B(A_j, A_j),$$

is a potential of the Killing metric h ;

(ii) *the moment map $\mu : G/K \rightarrow \mathfrak{k}^*$ associated with ρ is given by*

$$\mu(kaK)(X) = \frac{1}{2} \sum_{j=1}^r \sinh(2a_j) \hat{\rho}'(2a_j) B(\text{Ad}_{k^{-1}} C, K^j),$$

where $X \in \mathfrak{k}$.

Proof. (ii) Resume the notation of Proposition 3.1. It was shown in the proof Proposition 3.1 that for $z = aK$ one has

$$d^c \rho(\tilde{C}_z) = 0,$$

for all $C \in \mathfrak{m} \oplus \bigoplus_{\substack{\alpha \in \Sigma + \\ \alpha \neq 2e_j}} \mathfrak{k}[\alpha]$. Let $K^j \in \mathfrak{k}[2e_j]$ be defined as in (3). From (17) it follows that

$$d^c \rho(\tilde{K}^j_z) = -\frac{1}{2} \sinh(2a_j) \hat{\rho}'(2a_j) B(A_j, A_j). \quad (28)$$

As $B(A_j, A_j) = -B(K^j, K^j)$, then (28) and (14) imply (ii).

(i) Define $h_\rho(\cdot, \cdot) = -dd^c \rho(\cdot, I_0 \cdot)$. Because of the invariance of ρ and of the orthogonality relations proved in Proposition 3.1, it is sufficient to prove that $h_\rho(a_* P, a_* Q) = B(P, Q)$ for P, Q both in one of the blocks $a_* \mathfrak{a}$, $a_* \mathfrak{p}[e_j + e_l]$ and $a_* \mathfrak{p}[e_j]$.

The Hermitian form h_ρ on $a_* \mathfrak{a}$. Let $A_j, A_l \in \mathfrak{a}$, be as in (2). Then, by (13) and (9),

$$\begin{aligned} h_\rho(A_j, A_l) &= -dd^c \rho(a_* A_j, a_* I_0 A_l) = -dd^c \rho(a_* P^l, a_* A_j) \\ &= \frac{1}{\sinh(2a_l)} dd^c \rho((\tilde{K}^l)_z, (\tilde{A}_j)_z) = -\frac{1}{\sinh(2a_l)} \frac{d}{ds} \Big|_{s=0} \mu^{K^l}(\exp s A_j z), \end{aligned}$$

where $P^l \in \mathfrak{p}[2e_l]$ and $K^l \in \mathfrak{k}[2e_l]$ are defined in (3). By (ii) and (28) such quantity vanishes if $l \neq j$. For $j = l$, it becomes

$$\frac{1}{\sinh(2a_j)} \frac{1}{2} \frac{d}{ds} \Big|_{s=0} \sinh(2a_j + 2s) \hat{\rho}'(2a_j + 2s) B(A_l, A_l) =$$

$$\frac{1}{\sinh(2a_j)} (\cosh(2a_j) \hat{\rho}'(2a_j) + \sinh(2a_j) \hat{\rho}''(2a_j)) B(A_l, A_l) = B(A_l, A_l),$$

where the last equality follows from the assumption $\hat{\rho}'(t) = \frac{\cosh t - 1}{\sinh t}$.

Recall that if $\alpha \in \Sigma^+ \setminus \{2e_j\}$, then $I_0 \mathfrak{p}[\alpha] = \mathfrak{p}[\beta]$, for some $\beta \neq 0$. Let $P \in \mathfrak{p}[\alpha]$, with α as above. Write $P = X^\alpha - \theta X^\alpha \in \mathfrak{p}[\alpha]$, and $Q = I_0 P = X^\beta - \theta X^\beta \in \mathfrak{p}[\beta]$. Define $K := X^\alpha + \theta X^\alpha \in \mathfrak{k}[\alpha]$ and $C := X^\beta + \theta X^\beta \in \mathfrak{k}[\beta]$. As $K^j = [I_0 A_j, A_j]$, by (18) and (ii),

$$\begin{aligned} h_\rho(a_* P, a_* P) &= -\frac{1}{\sinh \alpha(H) \sinh \beta(H)} \frac{1}{2} \sum_k \sinh(2a_k) \hat{\rho}'(2a_k) B([C, K], [I_0 A_k, A_k]) = \\ &= -\frac{1}{\sinh \alpha(H) \sinh \beta(H)} \frac{1}{2} \sum_k \sinh(2a_k) \hat{\rho}'(2a_k) B(K, [[I_0 A_k, A_k], C]). \end{aligned}$$

From the Jacobi identity, one has

$$\begin{aligned} B(K, [[I_0 A_k, A_k], C]) &= -B(K, [[C, I_0 A_k], A_k] + [[A_k, C], I_0 A_k]) = \\ &= B([A_k, K], [I_0 A_k, C]) - B([I_0 A_k, K], [A_k, C]) = \\ &= \alpha(A_k) B(P, I_0 [A_k, C]) - \beta(A_k) B(I_0 [A_k, K], Q) = \\ &= (\alpha(A_k) \beta(A_k) + \beta(A_k) \alpha(A_k)) B(P, I_0 Q). \end{aligned}$$

As $I_0 Q = -P$, one obtains

$$h_\rho(a_* P, a_* P) = \frac{1}{2 \sinh \alpha(H) \sinh \beta(H)} \sum_k \sinh(2a_k) \hat{\rho}'(2a_k) (\alpha(A_k) \beta(A_k) + \beta(A_k) \alpha(A_k)) B(P, P).$$

We are left to check the following cases.

The Hermitian form h_ρ on $a_* \mathfrak{p}[e_j + e_l]$.

Here $\alpha = e_j + e_l$ and $\beta = e_j - e_l$. Then for $P \in \mathfrak{p}[e_j + e_l]$, one has

$$\begin{aligned} h_\rho(a_* P, a_* P) &= \frac{1}{2 \sinh(a_j + a_l) \sinh(a_j - a_l)} (\sinh(2a_j) \hat{\rho}'(2a_j) - \sinh(2a_l) \hat{\rho}'(2a_l)) B(P, P) = \\ &= \frac{\cosh(2a_j) - \cosh(2a_l)}{2 \sinh(a_j + a_l) \sinh(a_j - a_l)} B(P, P) = B(P, P), \end{aligned}$$

due to the identity $\cosh(2a_j) - \cosh(2a_l) = 2 \sinh(a_j + a_l) \sinh(a_j - a_l)$.

The Hermitian form h_ρ on $a_* \mathfrak{p}[e_j]$.

Here $\alpha = \beta = e_j$. Then for $P \in \mathfrak{p}[e_j]$, one has

$$\begin{aligned} h_\rho(a_* P, a_* P) &= \frac{1}{2 \sinh^2(a_j)} \sinh(2a_j) \hat{\rho}'(2a_j) B(P, P) \\ &= \frac{1}{2 \sinh^2(a_j)} (\cosh(2a_j) - 1) B(P, P) = B(P, P). \end{aligned}$$

This concludes the proof of (i). $\square$

The following remark shows that, up to an additive constant, the K -invariant potential ρ of the Killing metric h is unique.

Remark 5.2. Let ρ_1 and ρ_2 be smooth K -invariant functions on G/K such that $dd^c \rho_1 = dd^c \rho_2$. Then $\rho_1 - \rho_2$ is constant.

Proof. As $\rho_1 - \rho_2$ is pluriharmonic and G/K is contractible, there exists a unique holomorphic function $f : G/K \rightarrow \mathbb{C}$ such that $\operatorname{Re} f = \rho_1 - \rho_2$ (cf. [Gun90], Sect. K). By K -invariance, one has $\operatorname{Re} f = \operatorname{Re} f \circ k$, for every $k \in K$, and $f \circ k - f \equiv \lambda(k)$, where $\lambda(k)$ is a constant in $i\mathbb{R}$. Moreover, given $h, k \in K$ and $z \in G/K$, one has

$$f(z) + \lambda(hk) = f((hk) \cdot z) = f(k \cdot z) + \lambda(h) = f(z) + \lambda(h) + \lambda(k).$$

Hence the map $\lambda : K \rightarrow i\mathbb{R}$ is a Lie group homomorphism and it is necessarily trivial by the compactness of K . Thus f is K -invariant and it is also invariant with respect to the induced local $K^\mathbb{C}$ -action on G/K . Since $K^\mathbb{C}$ acts locally transitively on an open subset of G/K (cf. [Wol72]), the holomorphic function f is constant and so is its real part $\rho_1 - \rho_2$. $\square$

Since the logarithm of the Bergman kernel function is a K -invariant potential of the Killing metric (see [KoNo69], Vol.2, Exa. 6.6 p. 162 and Thm. 9.6 p. 262), one can draw the following conclusion.

Corollary 5.3. *Up to an additive constant, the smooth K -invariant exhaustion function ρ coincides with the logarithm of the Bergman kernel function.*

Example 5.4. *As an example, consider the unit disc $\Delta = G/K$, where $G = SU(1, 1)$ acts on Δ by linear fractional transformations. Fix the basis of $\mathfrak{g}$, normalized as in (3) :*

$$K^1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad P^1 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Then $\exp a_1 A_1 K = \tanh a_1 = |z|$. Choose $\hat{\rho}(t) = -\ln \frac{1}{\cosh t + 1}$, satisfying $\hat{\rho}'(t) = \frac{\cosh t - 1}{\sinh t}$. Since $B(A_1, A_1) = 8$, then up to an additive constant, the logarithm of the Bergman kernel function is given by

$$\begin{aligned} \rho(\exp a_1 A_1 K) &= -\frac{1}{4} \log \frac{1}{\cosh 2a_1 + 1} B(A_1, A_1) = \\ &= -2 \log \frac{\cosh^2 a_1 - \sinh^2 a_1}{2 \cosh^2 a_1} = -2 \log(1 - |z|^2) + \text{const}. \end{aligned}$$

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